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arXiv:

Keywords:

A Resonant Medium Framework for Cosmology: Finite-Memory Dynamics and the Echo Equation

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Abstract

Short abstract.

1 Introduction

Intro: context + prior art + hypothesis.

2 Framework

3 Datasets and Methods

Methods: data + runners + kernel fit.

4 Results

Results: Phase-8 spectral pass; Phase-20 robustness; Phase-21 guard-bands.

5 Discussion

Discussion: implications, predictions, limitations.

6 Conclusions

Data/Software/Code Availability

All figures and results are reproducible using the evidence bundle in `catalog/evidence/` (e.g., `echo_equation_evidence_bundle_v1.zip`).

AI-use statement

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References

- [1] Planck Collaboration, *Planck 2018 results. vi. cosmological parameters*, *Astronomy & Astrophysics* **641** (2020) A6, [[arXiv:1807.06209](#)].
- [2] D. Brout and D. e. a. Scolnic, *The pantheon+ analysis: Cosmological constraints*, *Astrophysical Journal* (2022).
- [3] DESI Collaboration, *Desi 2024 bao results from dr1*, *arXiv preprint* [[arXiv:2404.03000](#)].
- [4] J. Watts, “Echo equation: Resonant medium evidence summary.” Experimentalist Lab preprint, 2025.